



DiagramLens: Faceted Multimodal Retrieval for Diagrammatic Research Figures

Xinyue Chen¹

Yu Liu²

Chiteng Ma¹

Yuchu Luo¹

Xiaoru Yuan¹

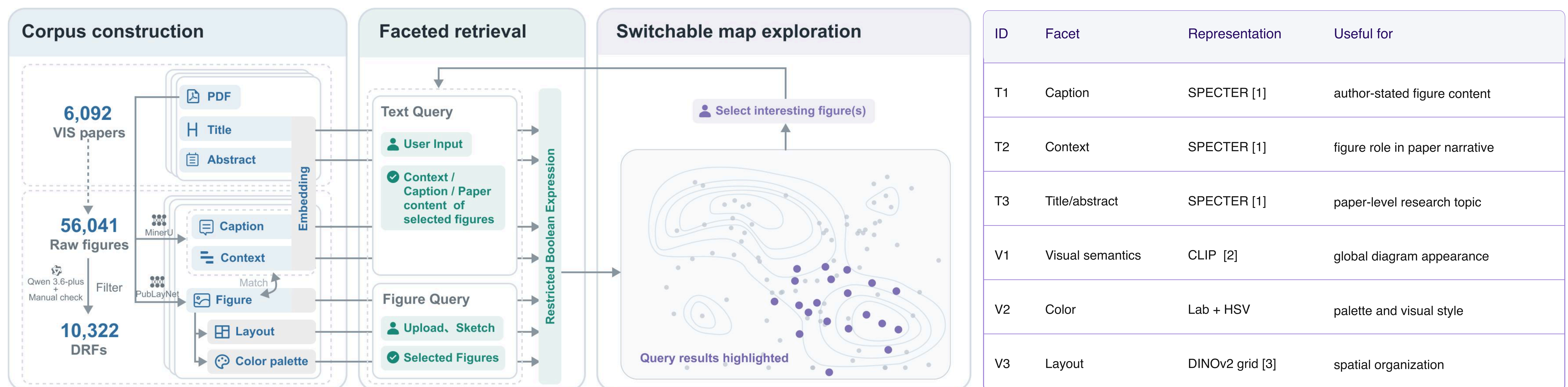
1) National Key Laboratory of General Artificial Intelligence and School of Intelligence Science and Technology, Peking University

2) University of Edinburgh, Edinburgh, United Kingdom.

Introduction

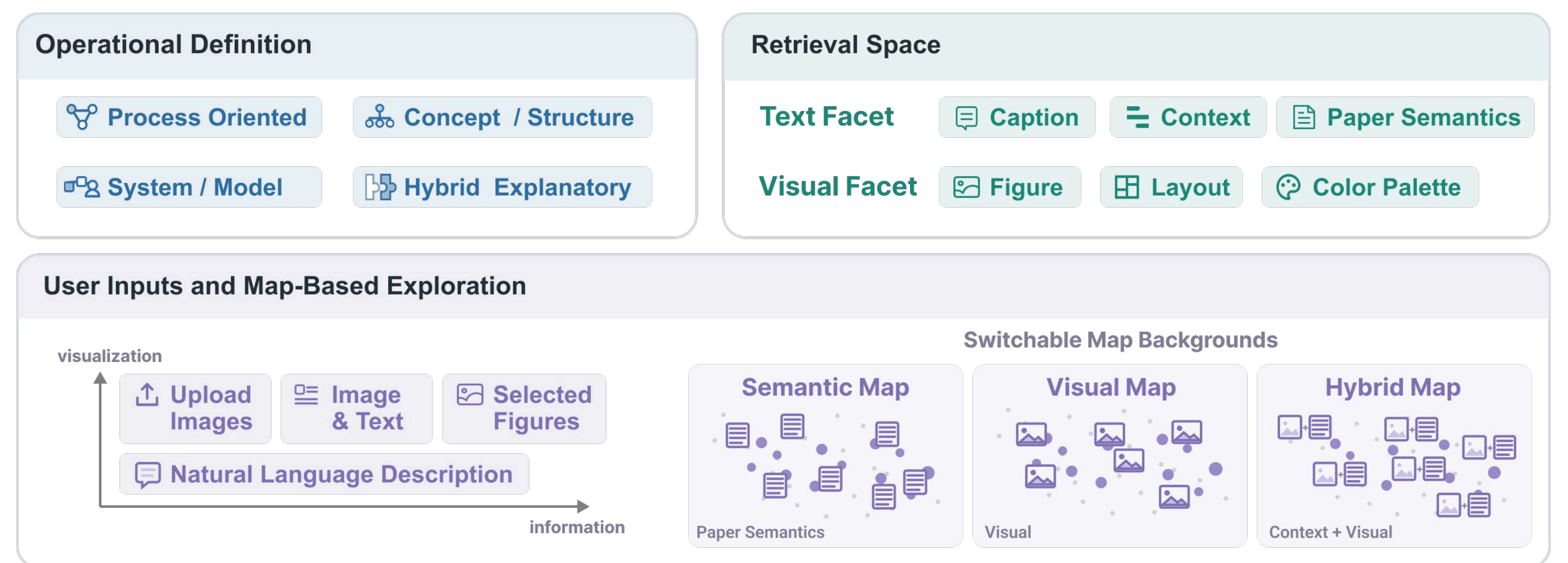
Diagrammatic research figures (DRFs), such as method overviews, system architectures, and taxonomies, are important entry points into visualization literature, but they are difficult to retrieve with text-only paper search or generic image retrieval. We present DiagramLens, a corpus and interactive retrieval prototype for faceted multimodal search over DRFs. Starting from 6,092 visualization papers, we construct a corpus of 10,322 DRFs and represent each figure through six complementary facets: caption, figure context, paper title/abstract, visual semantics, color, and layout. DiagramLens lets users combine these facets in the search space, inspect retrieved figures across semantic, visual, and hybrid projection maps, and interpret ranked results through facet-level explanations.

Corpus Construction and Retrieval Pipeline



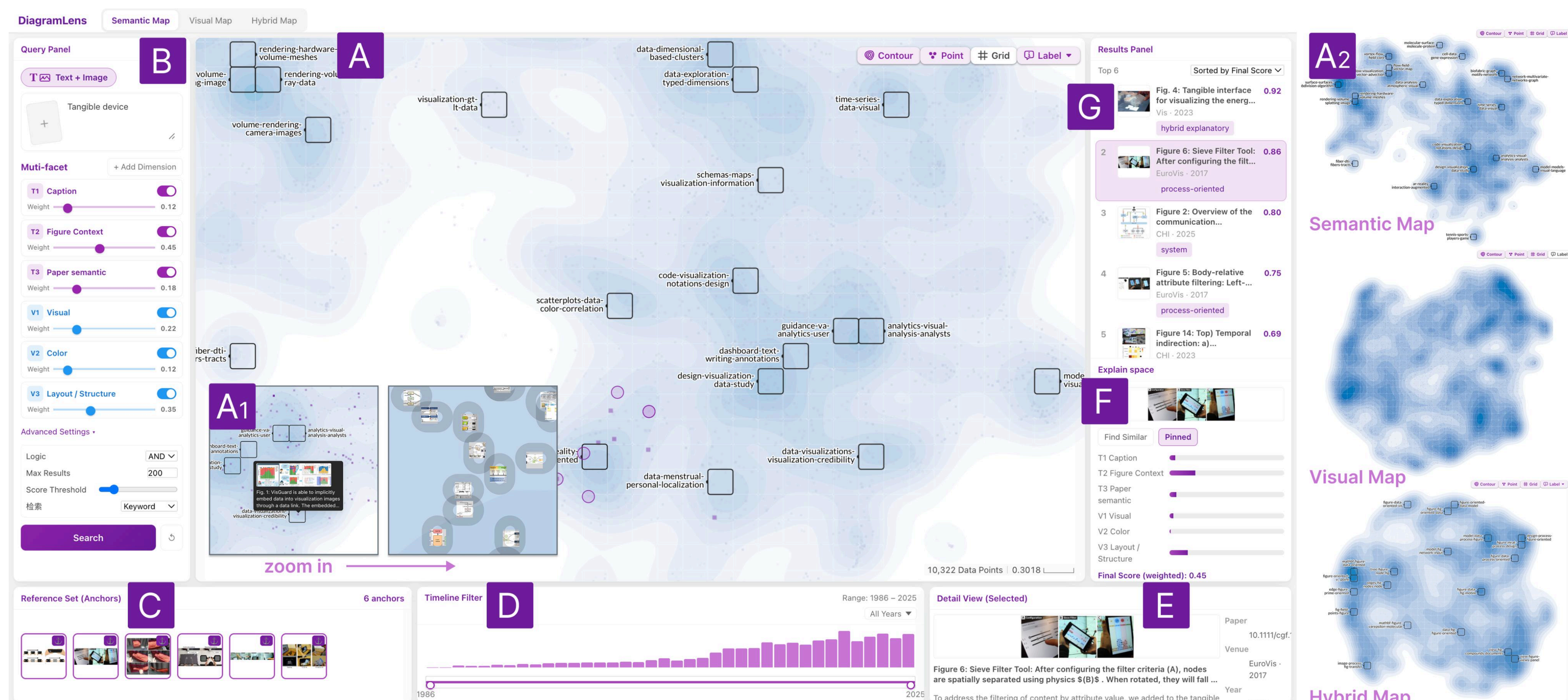
- We define diagrammatic research figures (DRFs) operationally and construct a corpus of 10,322 DRFs from 6,092 visualization papers and 56,041 raw figure images.

- We present **DiagramLens**, an interactive retrieval prototype for faceted multimodal search over DRFs. We propose a faceted retrieval model that represents each DRF through six complementary facets: caption, figure context, paper title/abstract, visual semantics, color, and layout.



System and Use Case

Interface

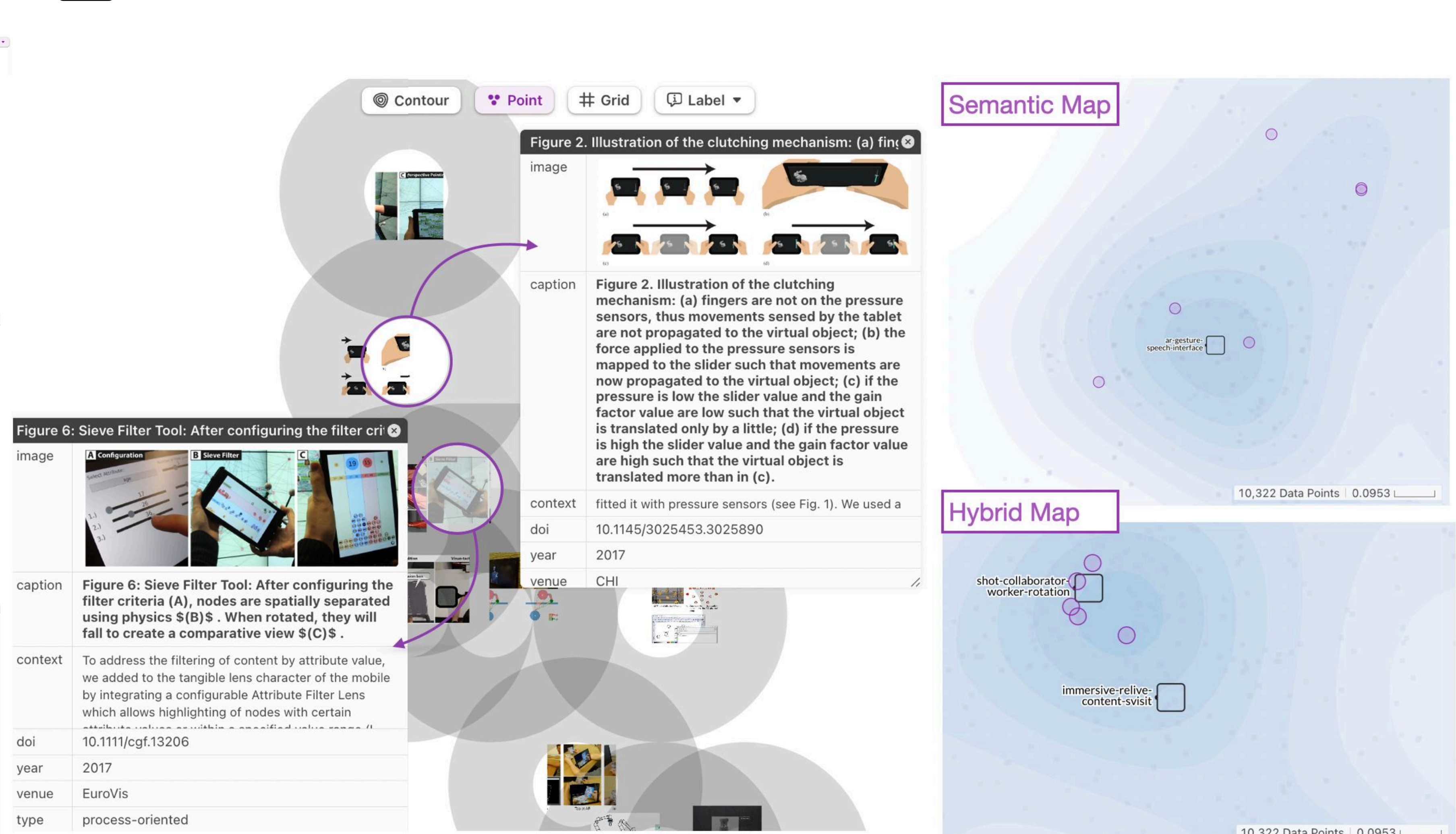


DiagramLens implements faceted DRF retrieval through three coordinated spaces. **The search space** supports natural-language queries, image queries, combined image-text queries, and selected figures as examples. **The projection space** shows the same retrieval results through semantic, visual, and hybrid maps, while **the explanation space** decomposes ranked results into per-facet contribution scores. The map view adopts a WizMap-style embedding-map view [4].

Reference

- [1] A. Cohan, S. Feldman, I. Beltagy, D. Downey, and D. S. Weld, "SPECTER: Document-level representation learning using citation-informed transformers," in Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2020, pp. 2270–2282.
- [2] A. Radford et al., "Learning transferable visual models from natural language supervision," in Proc. 38th Int. Conf. Mach. Learn. (ICML), 2021, pp. 8748–8763.
- [3] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," Trans. Mach. Learn. Res., 2024.
- [4] Z. J. Wang, F. Hohman, and D. H. Chau, "WizMap: Scalable interactive visualization for exploring large machine learning embeddings," in Proc. 61st Annu. Meeting Assoc. Comput. Linguistics (ACL), Volume 3: System Demonstrations, 2023, pp. 516–523.

Use Case



Starting from the query **"mobile interaction pipeline,"** the user retrieves DRFs related to mobile or handheld interaction and compares them through different map views. The semantic map keeps results organized by paper-topic context, while the hybrid map highlights cross-topic figures that share similar visual form. This helps users identify reusable diagrammatic patterns that are difficult to see from a ranked list alone.

Contact

xiaoru.yuan@pku.edu.cn
http://vis.pku.edu.cn

